TUTORIAL III

BIBLIOGRAPHY

III.1. INTRODUCTION

Bibliography is the environment which helps the author to cross-reference one publication from the list of sources at the end of the document. LaTeX helps authors to write a well structured bibliography, because this is how LaTeX works—by specifying structure.

It is easy to convert the style of bibliography to that of a publisher's requirement, without touching the code inside the bibliography. We can maintain a bibliographic data base using the program BibTeX. While preparing the articles, we can extract the needed references in the required style from this data base. harvard and natbib are widely used packages for generating bibliography.

To produce bibliography, we have the environment **thebibliography**[1], which is similar to the **enumerate** environment. Here we use the command `\bibitem` to separate the entries in the bibliography and use `\cite` to refer to a specific entry from this list in the document. This means that at the place of citation, it will produce number or author-year code connected with the list of references at the end.

```
\begin{thebibliography}{widest-label}
      \bibitem{key1}
      \bibitem{key2}
    \end{thebibliography}
```

The `\begin{thebibliography}` command requires an argument that indicates the width of the widest label in the bibliography. If you know you would have between 10 and 99 citations, you should start with

```
\begin{thebibliography}{99}
```

You can use any two digit number in the argument, since all numerals are of the same width. If you are using customized labels, put the longest label in argument, for example `\begin{thebibliography}{Long-name}`. Each entry in the environment should start with

```
\bibitem{key1}
```

If the author name is Alex and year 1991, the key can be coded as ale91 or some such mnemonic string[2]. This *key* is used to cite the publication within the document text. To cite a publication from the bibliography in the text, use the `\cite` command, which takes with the corresponding key as the argument. However, the argument to `\cite` can also be two or more keys, separated by commas.

[1] Bibiliography environment need two compilations. In the first compilation it will generate file with `aux` extension, where `citation` and `bibcite` will be marked and in the second compilation `\cite` will be replaced by numeral or author-year code.

[2] Key can be any sequence of letters, digits and punctuation characters, except that it may not contain a comma (maximum 256 characters).

`\cite{`*key1*`}` `\cite{`*key1,key2*`}`

In bibliography, numbering of the entries is generated automatically. You may also add a note to your citation, such as page number, chapter number etc. by using an optional argument to the `\cite` command. Whatever text appears in this argument will be placed within square brackets, after the label.

`\cite[page~25]{`*key1*`}`

See below an example of bibliography and citation. The following code

```
It is  hard to write unstructured and disorganised documents using
\LaTeX~\cite{les85}.It is interesting to typeset one
equation~\cite[Sec 3.3]{les85} rather than setting ten pages of
running matter~\cite{don89,rondon89}.

\begin{thebibliography}{9}
\bibitem{les85}Leslie Lamport, 1985. \emph{\LaTeX---A Document
Preparation System---User's Guide and Reference Manual},
Addision-Wesley, Reading.

\bibitem{don89}Donald E. Knuth, 1989. \emph{Typesetting Concrete
Mathematics}, TUGBoat, 10(1):31-36.

\bibitem{rondon89}Ronald L. Graham, Donald E. Knuth, and Ore
Patashnik, 1989. \emph{Concrete Mathematics: A Foundation for
Computer Science}, Addison-Wesley, Reading.
\end{thebibliography}
```

produces the following output:

It is hard to write unstructured and disorganised documents using LaTeX [1]. It is interesting to typeset one equation [1, Sec 3.3] rather than setting ten pages of running matter [2,3].

Bibliography

[1] Leslie Lamport, 1985. *LaTeX—A Document Preparation System—User's Guide and Reference Manual*, Addision-Wesley, Reading.

[2] Donald E. Knuth, 1989. *Typesetting Concrete Mathematics*, TUGBoat, 10(1):31-36.

[3] Ronald L. Graham, Donald E. Knuth, and Ore Patashnik, 1989. *Concrete Mathematics: A Foundation for Computer Science*, Addison-Wesley, Reading.

III.2. NATBIB

The natbib package is widely used for generating bibliography, because of its flexible interface for most of the available bibliographic styles. The natbib package is a re-implementation of the LaTeX `\cite` command, to work with both author–year and numerical citations. It is compatible with the standard bibliographic style files, such as plain.bst, as well as with those for harvard, apalike, chicago, astron, authordate, and of course natbib. To load the package; use the command.

`\usepackage[options]{natbib}`

III.2.1. Options for natbib

`round`	(default) for round parentheses
`square`	for square brackets
`curly`	for curly braces
`angle`	for angle brackets
`colon`	(default) to separate multiple citations with colons
`comma`	to use commas as separators
`authoryear`	(default) for author–year citations
`numbers`	for numerical citations
`super`	for superscripted numerical citations, as in *Nature*
`sort`	orders multiple citations into the sequence in which they appear in the list of references
`sort&compress`	as `sort` but in addition multiple numerical citations are compressed if possible (as 3–6, 15)
`longnamesfirst`	makes the first citation of any reference the equivalent of the starred variant (full author list) and subsequent citations normal (abbreviated list)
`sectionbib`	redefines `\thebibliography` to issue `\section*` instead of `\chapter*`; valid only for classes with a `\chapter` command; to be used with the `chapterbib` package
`nonamebreak`	keeps all the authors' names in a citation on one line; causes overfull hboxes but helps with some `hyperref` problems.

You can set references in the *Nature style* of citations (superscripts) as follows

```
\documentclass{article}
  \usepackage{natbib}
  \citestyle{nature}
\begin{document}
  . . . . . .
  . . . . . .
\end{document}
```

III.2.2. Basic commands

The natbib package has two basic citation commands, `\citet` and `\citep` for *textual* and *parenthetical* citations, respectively. There also exist the starred versions `\citet*` and `\citep*` that print the full author list, and not just the abbreviated one. All of these may take one or two optional arguments to add some text before and after the citation.

Normally we use author name and year for labeling the bibliography.

```
\begin{thebibliography}{widest-label}
  \bibitem{Leslie(1985)}{les85}Leslie Lamport, 1985.
    \emph{\LaTeX---A Document Preparation}...
  \bibitem{Donale(00)}{don89}Donald E. Knuth, 1989.
    \emph{Typesetting Concrete Mathematics},...
  \bibitem{Ronald, Donald and Ore(1989)}{rondon89}Ronald L. Graham, ...
\end{thebibliography}
```

Year in parentheses is mandatory in optional argument for bibitem. If year is missing in any of the bibitem, the whole author–year citation will be changed to numerical citation. To avoid this, give '(0000)' for year in optional argument and use partial citations (`\citeauthor`) in text.

Don't put 'space character' before opening bracket of year in optional argument.

```
\citet{ale91}                 ⇒  Alex et al. (1991)
\citet[chap.~4]{ale91}        ⇒  Alex et al. (1991, chap. 4)
\citep{ale91}                 ⇒  (Alex et al., 1991)
\citep[chap.~4]{ale91}        ⇒  (Alex et al., 1991, chap. 4)
\citep[see][]{ale91}          ⇒  (see Alex et al., 1991)
\citep[see][chap.~4]{jon91}   ⇒  (see Alex et al., 1991, chap. 4)
\citet*{ale91}                ⇒  Alex, Mathew, and Ravi (1991)
\citep*{ale91}                ⇒  (Alex, Mathew, and Ravi, 1991)
```

III.2.3. Multiple citations

Multiple citations may be made as usual, by including more than one citation key in the `\cite` command argument.

```
\citet{ale91,rav92}     ⇒  Alex et al. (1991); Ravi et al. (1992)
\citep{ale91,rav92}     ⇒  (Alex et al., 1991; Ravi et al. 1992)
\citep{ale91,ale92}     ⇒  (Alex et al., 1991, 1992)
\citep{ale91a,ale91b}   ⇒  (Alex et al., 1991a,b)
```

III.2.4. Numerical mode

These examples are for author–year citation mode. In numerical mode, the results are different.

```
\citet{ale91}                 ⇒  Alex et al. [5]
\citet[chap.~4]{ale91}        ⇒  Alex et al. [5, chap. 4]
\citep{ale91}                 ⇒  [5]
\citep[chap.~4]{ale91}        ⇒  [5, chap. 4]
\citep[see][]{ale91}          ⇒  [see 5]
\citep[see][chap.~4]{ale91}   ⇒  [see 5, chap. 4]
\citep{ale91a,ale91b}         ⇒  [5, 12]
```

III.2.5. Suppressed parentheses

As an alternative form of citation, `\citealt` is the same as `\citet` but *without any parentheses*. Similarly, `\citealp` is `\citep` with the parentheses turned off. Multiple references, notes, and the starred variants also exist.

```
\citealt{ale91}          ⇒  Alex et al. 1991
\citealt*{ale91}         ⇒  Alex, Mathew, and Ravi 1991
\citealp{ale91}          ⇒  Alex., 1991
\citealp*{ale91}         ⇒  Alex, Mathew, and Ravi, 1991
\citealp{ale91,ale92}    ⇒  Alex et al., 1991; Alex et al., 1992
\citealp[pg.~7]{ale91}   ⇒  Alex., 1991, pg. 7
\citetext{short comm.}   ⇒  (short comm.)
```

The `\citetext` command allows arbitrary text to be placed in the current citation parentheses. This may be used in combination with `\citealp`.

III.2.6. Partial citations

In author–year schemes, it is sometimes desirable to be able to refer to the authors without the year, or vice versa. This is provided with the extra commands

```
\citeauthor{ale91}     ⇒  Alex et al.
\citeauthor*{ale91}    ⇒  Alex, Mathew, and Ravi
\citeyear{ale91}       ⇒  1991
\citeyearpar{ale91}    ⇒  (1991)
```

III.2.7. Citations aliasing

Sometimes one wants to refer to a reference with a special designation, rather than by the authors, i.e. as Paper I, Paper II. Such aliases can be defined and used, textually and/or parenthetically with:

```
\defcitealias{jon90}{Paper~I}
```

```
\citetalias{ale91}   ⇒  Paper I
\citepalias{ale91}   ⇒  (Paper I)
```

These citation commands function much like `\citet` and `\citep`: they may take multiple keys in the argument, may contain notes, and are marked as hyperlinks.

III.2.8. Selecting citation style and punctuation

Use the command `\bibpunct` with one optional and six mandatory arguments:

1. The opening bracket symbol, `default = (`
2. The closing bracket symbol, `default = )`
3. The punctuation between multiple citations, `default = ;`
4. The letter 'n' for numerical style, or 's' for numerical superscript style, any other letter for author–year, `default = author--year;`
5. The punctuation that comes between the author names and the year
6. The punctuation that comes between years or numbers when common author lists are suppressed (`default = ,`);

The optional argument is the character preceding a post-note, default is a comma plus space. In redefining this character, one must include a space if that is what one wants.

Example 1

```
\bibpunct{[}{]}{,}{a}{}{;}
```

changes the output of

```
\citep{jon90,jon91,jam92}
```

into

[Jones et al. 1990; 1991, James et al. 1992].

Example 2

```
\bibpunct[;]{(}{)}{,}{a}{}{;}
```

changes the output of

```
\citep[and references therein]{jon90}
```

into

(Jones et al. 1990; and references therein).

TUTORIAL IV

BIBLIOGRAPHIC DATABASES

Bibliographic database is a database in which all the useful bibliographic entries can be stored. The information about the various publications is stored in one or more files with the extension `.bib`. For each publication, there is a *key* that identifies it and which may be used in the text document to refer to it. And this is available for all documents with a list of reference in the field. This database is useful for the authors/researchers who are constantly referring to the same publications in most of their works. This database system is possible with the BibTeX program supplied with the LaTeX package.

IV.1. The BibTeX program

BibTeX is an auxiliary program to LaTeX that automatically constructs a bibliography for a LaTeX document from one or more databases. To use BibTeX, you must include in your LaTeX input file a `\bibliography` command whose argument specifies one or more files that contain the database. For example

```
\bibliography{database1,database2}
```

The above command specifies that the bibliographic entries are obtained from *database1.bib* and *database2.bib*. To use BibTeX, your LaTeX input file must contain a `\bibliographystyle` command. This command specifies the *bibliography style*, which determines the format of the source list. For example, the command

```
\bibliographystyle{plain}
```

specifies that entries should be formatted as specified by the `plain` bibliography style (`plain.bst`). We can put `\bibliographystyle` command anywhere in the document after the `\begin{document}` command.

IV.2. BibTeX style files

plain Standard BibTeX style. Entries sorted alphabetically with numeric labels.

unsrt Standard BibTeX style. Similar to plain, but entries are printed in order of citation, rather than sorted. Numeric labels are used.

alpha Standard BibTeX style. Similar to plain, but the labels of the entries are formed from the author's name and the year of publication.

abbrv Standard BibTeX style. Similar to plain, but entries are more compact, since first names, month, and journal names are abbreviated.

acm Alternative BibTeX style, used for the journals of the Association for Computing Machinery. It has the author name (surname and first name) in small caps, and numbers as labels.

apalike Alternative BIBTEX style, used by the journals of the American Psychology Association. It should be used together with the LATEX apalike package. The bibliography entries are formatted alphabetically, last name first, each entry having a hanging indentation and no label.

Examples of some other style files are:

abbrv.bst, abstract.bst, acm.bst, agsm.bst, alpha.bst, amsalpha.bst, authordate*i*.bst, authordate1-4.sty, bbs.bst, cbe.bst, cell.bst, dcu.bst, harvard.sty, ieeetr.bst, jtb.bst, kluwer.bst, named.bst, named.sty, natbib.sty, natbib.bst, nature.sty, nature.bst, phcpc.bst, phiaea.bst, phjcp.bst, phrmp.bst plainyr.bst, siam.bst

Various organisations or individuals have developed style files that correspond to the house style of particular journals or editing houses. We can also customise a bibliography style, by making small changes to any of the `.bst` file, or else generate our own using the `makebst` program.

IV.2.1. Steps for running BIBTEX with LATEX

1. Run LATEX, which generates a list of `\cite` references in its auxiliary file, `.aux`.
2. Run BIBTEX, which reads the auxiliary file, looks up the references in a database (one or more `.bib` files, and then writes a file (the `.bbl` file) containing the formatted references according to the format specified in the style file (the `.bst` file). Warning and error messages are written to the log file (the `.blg` file). It should be noted that BIBTEX never reads the original LATEX source file.
3. Run LATEX again, which now reads the `.bbl` reference file.
4. Run LATEX a third time, resolving all references.

Occasionally the bibliography is to include publications that were *not* referenced in the text. These may be added with the command

```
\nocite{key}
```

given anywhere within the main document. It produces no text at all but simply informs BIBTEX that this reference is also to be put into the bibliography. With `\nocite{*}`, *every* entry in all the databases will be included, something that is useful when producing a list of all entries and their keys.

After running BIBTEX to make up the `.bbl` file, it is necessary to process LATEX *at least twice* to establish both the bibliography and the in-text reference labels. The bibliography will be printed where the `\bibliography` command is issued; it infact inputs the `.bbl` file.

IV.3. CREATING A BIBLIOGRAPHIC DATABASE

Though bibliographic database creation demands more work than typing up a list of references with the `thebibliography` environment; it has a great advantage that, the entries need to be included in the database only once and are then available for all future publications even if a different bibliography style is demanded in later works, all the information is already on hand in the database for BIBTEX to write a new `thebibliography` environment in another format. Given below is a specimen of an entry in bibliographic database:

```
@BOOK{knuth:86a,
 AUTHOR           ="Donald E. Knuth",
```

```
TITLE           ={The \TeX{}book},
EDITION         ="third"
PUBLISHER       ="Addison-Wesley",
ADDRESS         ={Reading, MA},
YEAR            =1986 }
```

The first word, prefixed `@`, determines the *entry_type*. The *entry_type* is followed by the reference information for that entry enclosed in curly braces { }. The very first entry is the *key* for the whole reference by which it is referred to in the `\cite` command. In the above example it is `knuth:86a`. The actual reference information is then entered in various *fields*, separated from one another by commas. Each *field* consists of a *field_name*, an = sign, with optional spaces on either side, and the *field text*. The *field_names* shows above are `AUTHOR`, `TITLE`, `PUBLISHER`, `ADDRESS`, and `YEAR`. The *field text* must be enclosed either in curly braces or in double quotation marks. However, if the text consists solely of a number, as for `YEAR` above, the braces or quotation marks may be left off.

For each entry type, certain fields are *required*, others are *optional*, and the rest are *ignored*. These are listed with the description of the various entry types below. If a required field is omitted, an error message will appear during the BIBTEX run. Optional fields will have their information included in the bibliography if they are present, but they need not be there. Ignored fields are useful for including extra information in the database that will not be output, such as a comment or an abstract of a paper. Ignored fields might also be ones that are used by other database programs.

The general syntax for entries in the bibliographic database reads

```
@entry_type{key,
  field_name = {field text},
  ....
  field_name = {field text} }
```

The names of the *entry_types* as well as the *field_names* may be written in capitals or lower case letters, or in a combination of both. Thus `@BOOK`, `@book`, and `@bOOk` are all acceptable variations.

The outermost pair of braces for the entire entry may be either curly braces { }, as illustrated, or parentheses ( ). In the latter case, the general syntax reads

```
@entry_type(key, ... ..)
```

However, the *field text* may only be enclosed within curly braces {...} or double quotation marks ... as shown in the example above.

The following is a list of the standard entry types in alphabetical order, with a brief description of the types of works for which they are applicable, together with the required and optional fields that they take.

`@article`:	Entry for an article from a journal or magazine.
required fields:	author, title, journal, year.
optional fields:	volume, number, pages, month, note.
`@book`:	Entry for a book with a definite publisher.
required fields:	author or editor, title, publisher, year.
optional fields:	volume or number, series, address, edition, month, note.
`@booklet`:	Entry for a printed and bound work without the name of a publisher or sponsoring organisation.
required fields:	title.
optional fields:	author, howpublished, address, month, year, note.

`@conference`:	Entry for an article in conference proceedings.
required fields:	author, title, booktitle, year.
optional fields:	editor, volume or number, series, pages, address, month, organisation, publisher, note.
`@inbook`:	Entry for a part (chapter, section, certain pages) of a book.
required fields:	author or editor, title, chapter and/or pages, publisher, year.
optional fields:	volume or number, series, type, address, edition, month, note.
`@incollection`:	Entry for part of a book that has its own title.
required fields:	author, title, booktitle, publisher, year.
optional fields:	editor, volume or number, series, type, chapter, pages, address, edition, month, note.
`@inproceedings`:	Entry for an article in conference proceedings.
required fields:	author, title, booktitle, year.
optional fields:	editor, volume or number, series, pages, address, month, organisation, publisher, note.
`@manual`:	Entry for technical documentation.
required fields:	title.
optional fields:	author, organisation, address, edition, month, year, note.
`@masterthesis`:	Entry for a Master's thesis.
required fields:	author, title, school, year.
optional fields:	type, address, month, note.
`@misc`:	Entry for a work that does not fit under any of the others.
required fields:	none.
optional fields:	author, title, howpublished, month, year, note.
`@phdthesis`:	Entry for a PhD thesis.
required fields:	author, title, school, year.
optional fields:	type, address, month, note.
`@proceedings`:	Entry for conference proceedings.
required fields:	title, year.
optional fields:	editor, volume or number, series, address, month, organisation, publisher, note.
`@unpublished`:	Entry for an unpublished work with an author and title.
required fields:	author, title, note.
optional fields:	month, year.

IV.3.1. Example of a LaTeX file (sample.tex) using bibliographical database (bsample.bib)

```
\documentclass{article}
\pagestyle{empty}
\begin{document}

\section*{Example of Citations of Kind \texttt{plain}}
Citation of a normal book~\cite{Eijkhout:1991} and an edited
book~\cite{Roth:postscript}. Now we cite an article written by a
single~\cite{Felici:1991} and by multiple
authors~\cite{Mittlebatch/Schoepf:1990}. A reference to an
article inside proceedings~\cite{Yannis:1991}.
We refer to a manual~\cite{Dynatext} and a technical
report~\cite{Knuth:WEB}. A citation of an unpublished
work~\cite{EVH:Office}. A reference to a chapter in a
book~\cite{Wood:color} and to a PhD thesis~\cite{Liang:1983}.
```

```
An example of multiple
citations~\cite{Eijkhout:1991,Roth:postscript}.

\bibliographystyle{plain} %% plain.bst
\bibliography{bsample}    %% bsample.bib
\end{document}
```

IV.3.2. Procedure for producing references for the above file sample.tex which uses bibliographic data base bsample.bib

```
$ latex sample     % 1st run of LaTeX

$ bibtex sample    % BibTeX run
                   % Then sample.bbl file will
                   % be produced

$ latex sample     % 2nd run of LaTeX
```

If still unresolved citation references

```
$ latex sample     % 3rd run of LaTeX
```